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Chapter 1

INTRODUCTION

Chapter 1 –INTRODUCTION

Introduction

Today we are embarking on a journey into future of public transportation with introduction of smart bus tickets. These are special tickets that make using the bus easier and better for everyone. With smart bus tickets, we won't need to worry about losing paper tickets or waiting in long lines. These digital tickets can be used on our phones or special cards, making bus travel faster and more convenient. In this presentation, we'll see how smart bus tickets work, why they are helpful, and how they can make our daily bus rides smoother and more enjoyable. Let's begin and learn about the future of bus travel with smart bus tickets.

As the population in urban areas continues to grow, there is often uncertainty regarding the arrival time of buses at stops. Urban bus networks are frequently complex and difficult to navigate, leading to further challenges. Additionally, buses are sometimes cancelled due to breakdowns, strikes, waterlogging, or other issues, leaving passengers uninformed about such disruptions [1]. The proposed Bus Ticketing System aims to address these challenges by providing an efficient framework that enables real-time bus tracking, estimated time of arrival (ETA) updates, ticket booking in advance, and maintaining a database of ticketing transactions [2][3]. These features are accessible through a flexible application available on the Android platform (or via a website) [4]. Real-time vehicle tracking is achieved using GPS, allowing users to monitor bus routes and locations. Users can view bus routes on a map, which includes both geographic and non-geographic details. GPS and Google Maps are utilized to display the current location of buses and related route information [5]. RFID tags are integrated to identify passengers and manage data such as payment and validity [6][7]. The system uses an ATmega328 microcontroller programmed with a Real-Time Clock (RTC) for efficient operations. Based on IoT, this system enables travellers to access bus information, such as routes and ETA, tailored to their source and destination through the Android application or website [8]. A key challenge of this system is providing all tracking, ETA, and ticketing functionalities directly to the user while ensuring

flexibility. Although these services could be offered at bus stops, doing so would defeat the purpose of creating a mobile and user-centric solution [9]. To address this, the Android application has been developed as a User Interface that simplifies access to all features and maximizes system functionality [10]. By offering real-time updates, route tracking, and a seamless ticketing experience, the Bus Ticketing System eliminates many of the issues caused by the unpredictability of bus arrivals, making public transportation more reliable and user-friendly.

Chapter 2

LITERATU REREVIEW

Chapter 2 – LITERATURE REVIEW

Literature Review

A literature review presents a concise overview of existing research related to the selected topic. It examines previously published studies to understand current developments, methodologies, and key findings. Reviewing earlier work helps identify research gaps and limitations in existing systems. This analysis provides a strong foundation and justification for the proposed study.

The paper “**Smart Bus Ticketing System**” by Divya R, Pavithra C V, Kamalabharathy A, Madhumeetha K, and Sowbarnikaa K, presented at the 2023 International Conference on Intelligent Technologies for Sustainable Electric and Communications Systems (iTech SECOM), proposes an automated and digital solution to modernize traditional bus fare collection systems. The main objective of the work is to overcome limitations of manual ticketing such as time consumption, human errors, revenue leakage, and inconvenience to passengers by introducing a smart, contactless ticketing approach. The system primarily uses **RFID technology** for passenger identification and ticket generation, where each passenger carries an RFID card that is scanned upon boarding the bus. Once scanned, the system automatically records travel details and deducts the appropriate fare without the need for paper tickets or manual intervention. To enhance monitoring and transparency, the system integrates **GPS** to track the real-time location of buses and **GSM communication** to transmit data, alerts, or confirmations to a centralized server or passengers. The entire setup operates under an **IoT framework**, enabling seamless data exchange between hardware components and backend systems for efficient fare management and transport monitoring. This approach significantly improves passenger convenience by enabling faster boarding, reduces operational costs, minimizes fraud, and provides transport authorities with accurate real-time data for analysis and planning. Overall, the paper highlights how the integration of RFID, GPS, GSM, and IoT technologies can create a reliable, efficient, and scalable smart bus ticketing system suitable for modern public transportation needs.

The paper “**Smart Digital Bus Ticketing System**” by **Nik Mohd Zarifie Hashim, Muhammad Azimi Bin Mohd Zarawi, Mohd Hariz Iswan Sinuin, and Mohd Khairuddin bin Abd Karim (2022)** presents a modern approach to public transportation ticketing by replacing conventional cash-based and paper-based systems with a **fully digital, cashless, and paperless solution**. The primary motivation of the study is to address common issues in traditional bus ticketing such as long boarding times, cash handling problems, ticket loss, fare manipulation, and lack of transaction transparency. The proposed system mainly utilizes **QR code technology and mobile applications**, allowing passengers to purchase tickets digitally using their smartphones and display a QR code for verification while boarding the bus. Bus conductors or automated scanners scan the QR code to validate the ticket, ensuring fast and secure passenger authentication. The system is supported by a backend database that records transaction details, passenger data, and journey information, which enhances accountability and reduces fraud. By integrating digital payment methods, the system eliminates the need for physical cash, improving safety and operational efficiency for both passengers and transport operators. Additionally, the use of mobile applications enables features such as ticket history tracking, real-time validation, and easy fare management. The authors emphasize that this digital ticketing approach improves passenger convenience, reduces operational costs, enhances data accuracy, and supports sustainable transportation by minimizing paper usage. Overall, the paper demonstrates that QR-code-based digital ticketing systems can significantly improve efficiency, security, and user experience in public bus transportation systems, making them suitable for smart city and intelligent transport applications.

The paper “**Smart Bus Pass System Using Android**” by **Mohammed Furkhan and H. R. Divakar (2022)** proposes a mobile-based solution to modernize the traditional bus pass system by shifting from manual, paper-based processes to a **digital, Android-driven platform**. The primary aim of the system is to simplify bus pass application, renewal, and verification while reducing administrative workload, paperwork, and processing time for transport authorities. In the proposed model, passengers use an **Android application** to register their personal details, select the type of bus pass (such as student, monthly, or employee pass), upload required documents, and submit applications online. Once approved by the transport authority through an admin interface, the pass is issued in digital form within the app, often embedded with a **QR code or unique digital ID** for validation during travel. Conductors or inspectors verify

the pass by scanning the QR code using a mobile device, ensuring quick and secure authentication without physical passes. All passenger and transaction data are stored in a centralized database, enabling efficient record management, easy renewal tracking, and reduced chances of pass duplication or misuse. The system also supports online pass renewal and digital payment options, making the process convenient and time-saving for users. Overall, the paper highlights that an Android-based smart bus pass system enhances passenger convenience, improves operational efficiency, minimizes fraud, and supports sustainable public transport by eliminating paper passes and manual verification processes.

The paper **“IoT Based Efficient Contactless Bus Ticketing System”** by **Pulla Rushika, P. R. Yashaswini, Peravali Sownya Sri, Soundarya Shivakumar, Swamy Hiremath, and Roopa R. Kulkarni (2021)** presents a modern public transportation solution that replaces traditional cash-based and manual ticketing with an **IoT-enabled, contactless fare collection system**. The proposed system uses **smart cards** assigned to passengers, which are scanned when boarding the bus to automatically identify the user and deduct the appropriate fare without physical interaction. Through **real-time data processing and IoT connectivity**, ticketing and travel information is transmitted to a centralized server, allowing transport authorities to monitor transactions, manage revenue, and analyze travel patterns efficiently. This approach significantly reduces boarding time, human errors, and ticket fraud while minimizing physical cash handling, thereby improving safety and passenger convenience. Overall, the system enhances operational efficiency, transparency, and reliability of public transport services, making it suitable for smart city and intelligent transportation applications.

The paper **“Smart Bus: A Smartphone-based Framework for Public Transport Ticket System using QR Code and Implementing Delay Payment Method”** by **Ajinkya Gaikwad (2021)** proposes an innovative mobile-centric ticketing solution designed to improve convenience and flexibility in public transportation. In this system, passengers use a **smartphone application** to generate digital bus tickets in the form of **QR codes**, which are scanned by conductors or automated devices for quick and secure validation during boarding. A key contribution of the work is the **delay payment mechanism**, which allows passengers to travel even when immediate payment is not possible, with the fare being settled later through the app, thereby addressing issues such as low balance or network unavailability at the time of travel. **The system supports**

cashless transactions, reducing the need for physical money handling and paper tickets while improving boarding speed and operational efficiency. Ticket and payment details are stored in a backend database, ensuring transparency, accountability, and easy tracking of travel history. Overall, the paper demonstrates that a QR-code-based smartphone ticketing system with delayed payment capability can enhance passenger convenience, reduce fare collection issues, and support smarter, more flexible public transport systems.

The paper “**Design of Smart Bus Ticketing System using IoT and RFID**” by **Shaik Saidu Mashan and Dr. Christeena Joseph (2020)** proposes an automated and intelligent approach to replace traditional manual bus ticketing systems. The system is based on the use of **RFID cards** issued to passengers, which are scanned when boarding the bus to uniquely identify each traveler and automatically record journey details. Once the RFID card is detected, the system calculates and deducts the appropriate fare without the need for paper tickets or cash transactions. By integrating the RFID-based ticketing mechanism with an **IoT framework**, all ticketing and passenger data are transmitted in real time to a centralized server, enabling effective data storage, monitoring, and management by transport authorities. This real-time connectivity improves transparency, enhances security by reducing ticket fraud and duplication, and minimizes human errors commonly associated with manual ticket issuance. Additionally, the system reduces boarding time and improves passenger convenience through faster, contactless transactions. From an operational perspective, transport operators benefit from improved revenue tracking, efficient fleet management, and access to accurate travel data for analysis and planning. Overall, the paper highlights that the integration of IoT and RFID technologies can significantly enhance the efficiency, reliability, and user experience of public transportation ticketing systems.

The paper “**NFC Based Intelligent Bus Ticketing System**” by **Rajshree Hargude, Aarti Kamthe, and Prof. Vinod Badgujar (2020)** presents a modern and secure approach to bus fare collection by adopting **Near Field Communication (NFC) technology** for contactless ticketing. In the proposed system, passengers are provided with **NFC-enabled cards or smartphones**, which are tapped on an NFC reader installed inside the bus to automatically identify the passenger and initiate the ticketing process. Once the NFC tag is read, the system records travel details and deducts the

appropriate fare digitally, eliminating the need for cash transactions and paper tickets. The system is connected to a **centralized backend database**, enabling real-time data transmission, storage, and monitoring of ticketing transactions and passenger information. This real-time tracking helps transport authorities in effective revenue management, passenger flow analysis, and route optimization. By minimizing manual intervention, the system significantly reduces human errors, ticket fraud, and boarding delays. Additionally, the NFC-based approach enhances passenger convenience by offering faster boarding, secure cashless payments, and a seamless travel experience. Overall, the paper demonstrates that NFC technology can play a vital role in developing efficient, reliable, and intelligent public transportation systems suitable for smart city applications.

The paper “**Smart Bus Monitoring and Ticketing System Using IoT**” by **Shanmugapriya R, Sowmya A, Sowmya S, and Shriram S R (2020)** presents an integrated **IoT-based solution** that combines bus monitoring and automated ticketing to improve the efficiency and sustainability of public transportation systems. The proposed system uses **GPS technology** to track buses in real time, allowing passengers and transport authorities to monitor bus locations, schedules, and delays accurately. **RFID technology** is employed for automated fare collection, where passengers carry RFID cards that are scanned upon boarding, enabling contactless ticket generation and digital fare deduction without manual intervention. All ticketing, location, and passenger data are transmitted to a **cloud-based server**, ensuring centralized data storage, real-time monitoring, and easy access for transport administrators. This integration reduces boarding time, minimizes ticketing errors and fraud, and enhances operational transparency. Additionally, by improving route efficiency, reducing idle time, and encouraging the use of public transport through better service quality, the system contributes to **lower fuel consumption and reduced greenhouse gas emissions**. Overall, the paper highlights how IoT, GPS, RFID, and cloud technologies can work together to create a smart, efficient, and environmentally friendly public transportation monitoring and ticketing system.

The paper “**RFID Based Smart Ticketing System for Public Transportation**” by **Shreya Andhale, Nirmayee Dighe, Akanksha Kore, Darshana Gaikwad, and Yogesh Deshmukh (2020)** proposes an automated fare collection system that replaces traditional manual ticketing with an **RFID-based, cashless solution** to improve public

transport efficiency and passenger convenience. In the proposed system, passengers are issued **RFID cards** that store unique identification information and prepaid balance details. When a passenger boards a bus, the RFID card is scanned by a reader installed inside the vehicle, which automatically identifies the passenger, records the journey, and deducts the appropriate fare without the need for cash or paper tickets. All transaction and travel data are stored in a **centralized database**, enabling accurate record keeping, easy fare management, and real-time monitoring by transport authorities. By eliminating manual ticket issuance, the system significantly reduces human errors, ticket fraud, and boarding delays while improving transparency and revenue control. Additionally, the RFID-based approach enhances passenger comfort by enabling faster boarding and seamless travel. Overall, the paper highlights that RFID technology can play a crucial role in creating efficient, reliable, and user-friendly smart ticketing systems for modern public transportation networks.

The paper “**Smart Bus Ticket System Using IoT Technology**” by **Sumit Kumar, Piyush Kumar, Kartikay Tyagi, and Ayush Kumar (2019)** proposes an **IoT-based automated bus ticketing system** aimed at replacing traditional manual fare collection methods with a smart, efficient, and cashless solution. In the proposed system, passengers use **smart cards** that are scanned when boarding the bus to automatically identify the traveler and record journey details. Once the card is detected, the fare is calculated and deducted digitally without the involvement of conductors or paper tickets. The system uses **real-time data processing through IoT connectivity**, enabling ticketing information and transaction data to be transmitted instantly to a centralized server for storage and monitoring. This real-time data access helps transport authorities manage revenue effectively, monitor passenger flow, and analyze travel patterns. By reducing manual intervention, the system minimizes human errors, prevents fare manipulation, and speeds up the boarding process, thereby improving overall operational efficiency. Additionally, the contactless and automated nature of the system enhances passenger convenience by offering faster, safer, and seamless travel. Overall, the paper demonstrates how IoT-enabled smart ticketing systems can significantly improve reliability, transparency, and efficiency in modern public transportation networks.

The paper “**A Review on Smart Bus Ticketing and Tracking System using IoT**” by **Bhat Apoorva, Kavitha C, Bharath M, Nitesh Narayan Vaidya, and Prof.**

Harshitha G M (2019) provides a comprehensive survey of existing **IoT-based smart bus ticketing and tracking solutions** aimed at improving the efficiency, reliability, and convenience of public transportation systems. The authors review various technologies such as **RFID, GPS, GSM, NFC, QR codes, and mobile applications** that are commonly used for automated fare collection and real-time vehicle tracking. The paper explains how RFID and smart cards enable contactless and cashless ticketing, while GPS and GSM technologies support real-time bus location tracking and communication with centralized servers. It also discusses the role of cloud platforms and IoT frameworks in enabling data storage, monitoring, and analytics for transport authorities. In addition to highlighting the benefits—such as reduced manual errors, faster boarding, improved passenger experience, better revenue management, and enhanced operational transparency—the paper also addresses key challenges in implementation, including infrastructure cost, network reliability, data security, scalability, and maintenance. Overall, the review emphasizes that IoT-based smart bus ticketing and tracking systems have strong potential to support intelligent transportation and smart city initiatives, while careful system design is necessary to overcome practical deployment challenges.

The paper “**Online Bus Tracking and Ticketing System**” by **Dipika Tidke, Vinayak Nayar, Amit Pawar, and Vishakha Pagar (2018)** proposes a web-based solution that integrates **real-time bus tracking and online ticket booking** to improve the efficiency and usability of public transportation systems. The system uses **GPS technology** installed in buses to continuously track their real-time location, which is transmitted to a central server and made available to passengers through an online platform or mobile interface. This allows users to check bus arrival times, routes, and current positions, thereby reducing uncertainty and waiting time at bus stops. In addition to tracking, the system enables **online ticket booking**, allowing passengers to reserve or purchase tickets in advance without the need to stand in queues or handle cash. All booking, tracking, and passenger data are stored in a centralized database, ensuring accurate record management and smooth coordination between users and transport operators. By combining GPS-based tracking with online ticketing, the system improves passenger convenience, reduces manual workload, enhances transparency, and streamlines transport operations. Overall, the paper highlights how online and GPS-enabled solutions can significantly enhance public transport services by making them more reliable, user-friendly, and efficient.

The paper “**RFID-based Smart Bus using Embedded System**” by **Mr. D. Baskaran, Mr. M. Pattumuthu, Ms. B. Priyadharshini, Mr. P. Shabab Akram, and Ms. S. Sripriya (2016)** presents an early implementation of a smart public transportation system that uses **embedded systems and RFID technology** to automate bus ticketing and passenger identification. In the proposed system, passengers are issued **RFID tags or cards** that store unique identification details and prepaid balance information. When a passenger boards the bus, the RFID reader connected to an embedded controller scans the card, verifies the passenger’s identity, and automatically deducts the appropriate fare without the need for manual ticket issuance or cash handling. The embedded system processes the transaction in real time and can display ticketing information on an onboard display unit. This approach significantly reduces human errors, boarding time, and ticket fraud while enhancing security through authenticated access. Additionally, the system improves passenger convenience by enabling **contactless and cashless travel**, and it supports better revenue management for transport authorities. Overall, the paper demonstrates how RFID integrated with embedded systems can effectively modernize bus transportation by improving efficiency, reliability, and user experience.

The paper “**Real Time Web Based Bus Tracking**” by **Manini Kumbhar, Meghana Survase, Pratibha Mastud, and Avadhut Sakunkhe (2016)** presents a **web-based solution designed to improve public transportation efficiency and commuter convenience through real-time bus location tracking**. The proposed system uses **GPS modules installed in buses** to continuously capture **location coordinates**, which are transmitted to a central server using communication networks. These live location updates are then displayed on a **web-based platform**, allowing passengers to track buses in **real time** and obtain **accurate information about bus arrival times and routes**. By providing instant access to bus movement data, the system significantly reduces passenger waiting time and uncertainty at bus stops. Transport authorities also benefit from improved fleet monitoring, route management, and schedule adherence. The paper highlights that integrating GPS with web technologies enhances transparency, improves operational efficiency, and encourages greater use of public transportation by offering reliable and user-friendly tracking services.

The paper “**GPS Supported Bus Tracking and Ticketing System**” by **Ajay Shingare, Ankita Pendole, and Nikita Choudhary (2015)** proposes an **integrated public**

transportation solution that combines **real-time bus tracking** with a **digital ticketing system** to enhance overall transport efficiency and passenger convenience. In the proposed system, **GPS technology** is installed in buses to continuously monitor and transmit their real-time location to a central server, allowing passengers to track bus movements and estimate arrival times accurately. Alongside tracking, the system incorporates a **smart ticketing mechanism**, where fare collection is handled digitally instead of through manual, paper-based methods. This reduces the dependence on cash transactions, minimizes human errors, and speeds up the ticketing process. All tracking and ticketing data are stored in a centralized database, enabling transport authorities to monitor fleet performance, manage revenue, and analyze travel patterns effectively. By integrating GPS-based monitoring with automated ticketing, the system improves service reliability, reduces passenger waiting time, and provides a more transparent and efficient public transportation experience.

The paper “**Mobile Enabled Bus Tracking and Ticketing System**” by **Suresh Shankar Narayan and Paul Hamilton (May 2014)** presents a mobile-centric solution that integrates **GPS and RFID technologies** to improve the efficiency and usability of public transportation systems. In the proposed system, **GPS modules** installed in buses continuously track their real-time location, and this information is transmitted to a central server, allowing passengers to view live bus locations and estimated arrival times through a **mobile application**. For ticketing, the system uses **RFID-based digital fare collection**, where passengers carry RFID cards that are scanned when boarding the bus to automatically identify the user and deduct the fare without the need for cash or paper tickets. All tracking and ticketing data are stored in a centralized database, enabling transport authorities to monitor bus movement, passenger usage, and revenue in real time. This integration reduces manual intervention, minimizes ticketing errors and fraud, speeds up boarding, and enhances passenger convenience through real-time information access and cashless transactions. Overall, the paper demonstrates how combining mobile technology with GPS and RFID can create a reliable, efficient, and user-friendly smart transportation system suitable for modern public transit operations.

Chapter 3

GAP ANALYSIS

Chapter 3 – GAP ANALYSIS

Gap Analysis

The existing public bus transport systems have the capacity to absorb large masses of urban travellers, their public image often suffers from a negative perception. From a passenger's point of view, bus networks in dense urban areas are often considered as complex and difficult to navigate. In contrast to private modes of transport, traveling on busses offers only a low level of comfort and convenience. Bus journeys lack a sense of personal control and ownership that is valued by car users.

The 2014 “Mobile Enabled Bus Tracking and Ticketing System” had limitations in providing accurate bus location and timing. These limitations can be addressed by integrating GPS with microcontrollers like ATmega328 and ESP32, enabling real-time tracking and more precise location updates.

The 2015 “GPS Supported City Bus Tracking and Ticketing System” faced limitations when network connectivity was unavailable, affecting real-time tracking. This limitation can be addressed by implementing offline data handling or local storage to maintain system operation during network issues.

The 2016 “Real-Time Web-Based Bus Tracking” system had limitations in accuracy and lacked cloud integration. These issues can be addressed by using cloud storage for real-time data management, which improves reliability and provides more accurate tracking results.

The 2018 “Online Bus Tracking and Ticketing System” required manual ticket punching, which was time-consuming. This limitation can be overcome by using RFID cards and readers, enabling faster and automated fare collection.

The 2020 “Smart Bus Monitoring and Ticketing System Using IoT” did not record passenger travel history. This limitation can be addressed by storing passenger travel data, enabling better tracking and analysis of usage patterns.

The 2022 “Smart Bus Ticketing System” used QR codes, which sometimes caused issues. This can be improved by using RFID cards and readers, allowing faster transactions and easy balance recharging via the conductor's machine.

The 2021 “RFID-Based Smart Bus Ticketing for Public Transportation” used NodeMCU, which has limited pins and memory. This limitation can be addressed by using ESP32 and ATmega328 for better performance and flexibility.

The 2022 “Smart Ticketing System for Public Transport” did not provide real-time data. This limitation can be overcome by integrating GPS and cloud storage to enable real-time tracking and updates.

The 2022 “Proposed System on Smart Bus Ticketing” used GTFS, which does not provide real-time updates. This limitation can be overcome by using the MQTT protocol for real-time communication.

Chapter4

PROBLEM STATEMENT

Chapter 4 – PROBLEM STATEMENT

Problem Statement

- Passengers might lose the printed tickets purchased from the counter.
- Passengers have to go to the ticket counter to purchase tickets and it is not very convenient as it takes much time and effort.
- In manual System, staff need to print the ticket during purchasing process at counter. This will require a relatively high cost to buy paper to print the tickets.
- To develop an easy way of ticket management System which help to shorten the queue and passengers stress for getting a ticket and passengers convenience in general.

Chapter 5

OBJECTIVE

Chapter 5 – OBJECTIVE

Objective

- **Reduce conductor efforts-** Using this system we can reduce the efforts which have been taken by the conductor while using the traditional system.
- **Improve work quality-** Using this system we can improve work quality and increase efficiency of work.
- **Create passenger records for MSRTC on Server-** The RFID number and all detail information of a every passenger as well as passenger history has been recorded on MSRTC server.
- **Analysing of passenger frequency to particular route -** Using this system we can easily understand the total number of passengers in particular route.
- **Bring complete digitalization into government transport service-** As we know our India is on the way of complete digitalization so it's one step from our side to bring digitalization in transport system.
- **Avoid frauds and loss-** Using this system we can find and avoid frauds and loss which present in transport system, people cannot take ticket while travelling so it's one of the losses for transport system we can avoid it.

Chapter 6

PROPOSED METHODOLOGY

Chapter 6-PROPOSED METHODOLOGY

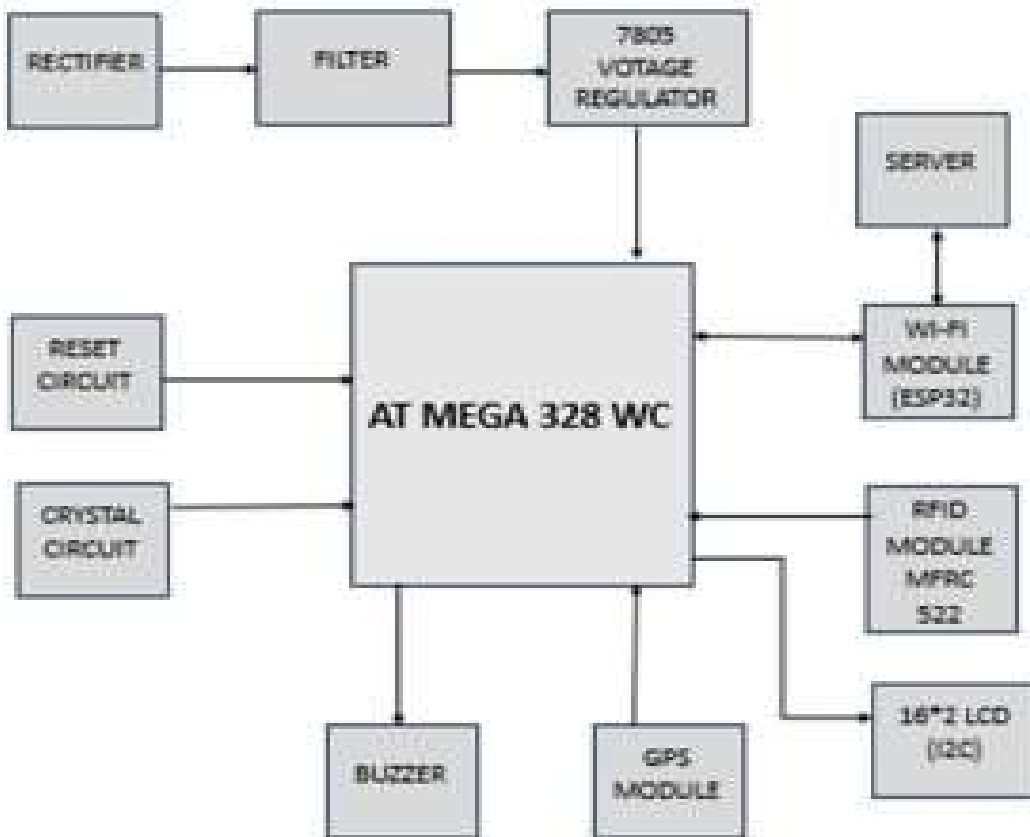
6.1 Proposed Methodology

The proposed methodology integrates RFID and GPS technologies to automate bus fare calculation and improve route efficiency. Each bus is equipped with an RFID reader and a GPS module connected to an onboard processing unit. When a passenger boards the bus, they tap their RFID card, and the system records the card ID along with the boarding time and GPS coordinates. The passenger's trip is temporarily registered as active. Upon alighting, the passenger taps the card again, and the exit time and GPS coordinates are captured. The distance traveled is then computed using the Haversine formula based on the entry and exit GPS coordinates. This calculated distance is multiplied by a predefined per-kilometer rate to determine the fare. If the passenger's card balance is sufficient, the fare is automatically deducted and the updated balance is stored in the central database; otherwise, the system generates an insufficient balance alert.

All transaction and location data are transmitted to a centralized server in real time, where it is stored and analyzed. The collected data is further used to monitor bus occupancy, passenger travel patterns, and revenue generation. Using historical GPS and RFID logs, the system performs route analysis to identify high-demand segments and peak travel periods. These insights support route optimization, improved scheduling, and better fleet utilization. Overall, the methodology ensures faster boarding, accurate fare calculation, reduced manual intervention, and data-driven decision-making for efficient public transportation management.

6.2 Block Diagram and Working

Block diagram-



Working-

The supply comes from an adapter it goes to the voltage regulator and then to the microcontroller, and if from transformer then the supply gives to rectifier to convert AC into DC. This process is called rectification. The output of the rectifier is DC, but it is not pure DC. To obtain pure DC, we use a filter. The voltage at this stage is typically 12V, but microcontrollers require only 5V. For this conversion, we use a 7805-voltage regulator. A reset circuit is used as a functional switch. A resistor is connected across the reset switch to ground, and a second resistor is placed across the LED with its negative terminal. If we need to restart the microcontroller, the reset circuit allows us to do so. The crystal circuit is used to provide oscillation to the microcontroller. A crystal oscillator is placed across two ceramic capacitors

to achieve this. For interfacing an RFID reader, we use an SPI (Serial Peripheral Interface) communicator.

We use four communication protocols as follows-

1. To interface microcontroller and RFID reader we use SPI communication protocol.
SPI (Serial Peripheral Interface) communication is used in a smart bus ticketing system to transfer data between devices like smart cards, card readers, microcontrollers, and displays.
-Card Reader and Smart Card: When a passenger taps their smart card on the card reader, the reader uses SPI communication to exchange data with the smart card. The card sends information like the balance or ticket details to the reader.
-Microcontroller as a Controller: The card reader is connected to a microcontroller (the "brain" of the system) via SPI. The microcontroller processes the data from the card, checks the balance, and deducts the fare.
2. To interface server and WIFI module we use MQTT communication protocol.
MQTT ensures smooth, real-time communication between all parts of a smart bus ticketing system, aiding operations like ticket sales, route tracking, and more. It uses very little data and works well even with a weak internet connection, making it suitable for buses traveling through areas with poor network coverage. This makes the system fast, efficient, and reliable.
3. To interface GPS, we use UART communication protocol.
UART (Universal Asynchronous Receiver-Transmitter) communication is used to interface a GPS module in a smart bus ticketing system to track the bus location. Here's how it works in simple terms: Connecting the GPS Module: The GPS module, which determines the bus's location, communicates with the microcontroller (the system's brain) using UART. This involves two wires: TX (Transmit): Sends data from the GPS module to the microcontroller. RX (Receive): Receives data from the microcontroller (if needed).
4. To interface LCD, we use I2C Communication protocol.
I2C (Inter-Integrated Circuit) communication is widely used in smart bus ticketing systems due to its simplicity, efficiency, and ability to connect multiple devices over just two wires (SDA for data and SCL for clock). Here's how I2C communication is utilized in a smart bus ticket system: Components in the System:
-RFID Reader/Tag Module: Used to scan passenger cards or tickets.
Communicates with the microcontroller via

-Display Module (LCD/OLED) Shows fare details, remaining balance, or confirmation messages. Communicates with the microcontroller via I2c

Working of I2C in a Smart Bus Ticketing System: Passenger Authentication:

When a passenger taps their RFID card on the reader, the reader communicates the unique ID of the card to the microcontroller via I2C.

6.3 Flow Chart

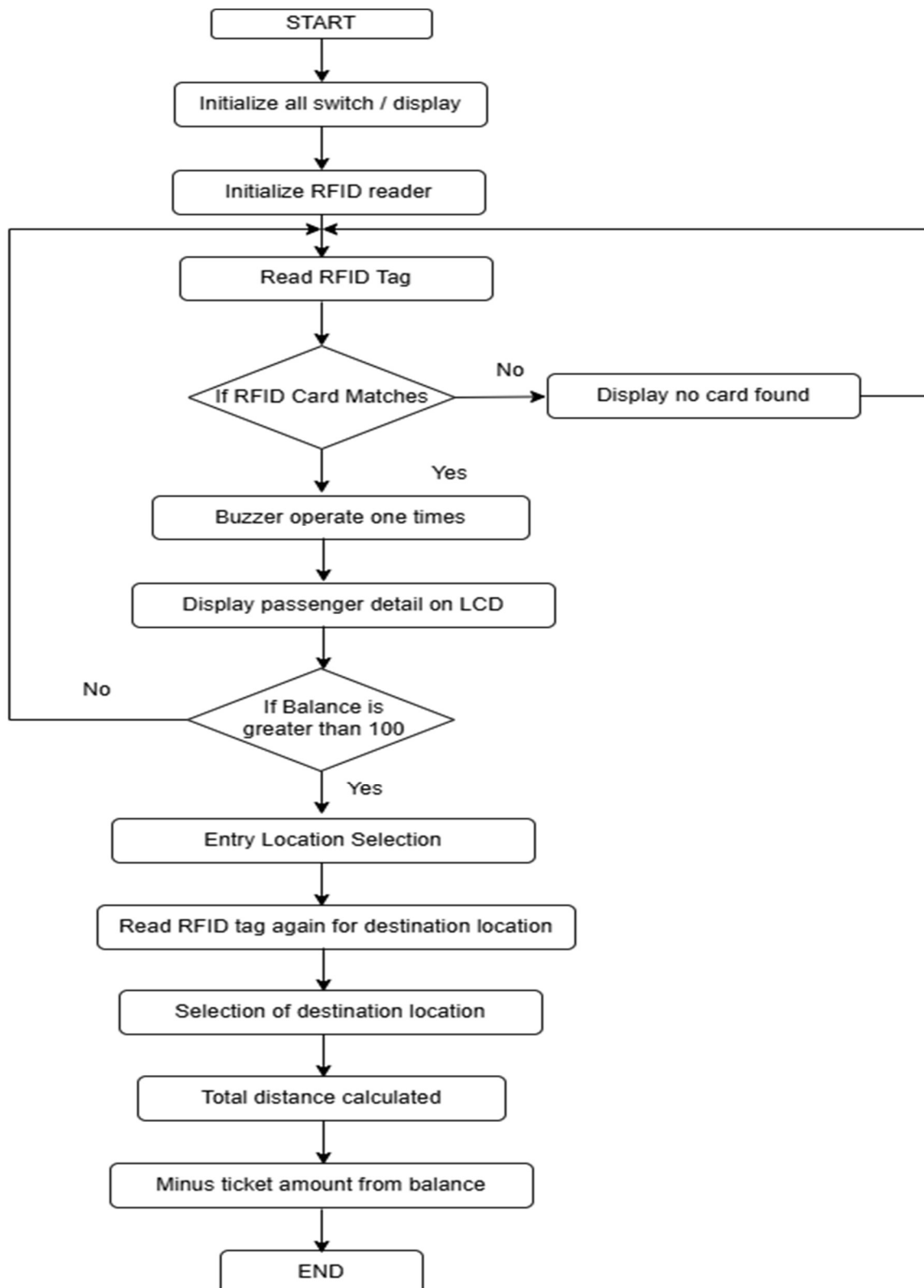


Figure 2 : Flow chart

The flowchart outlines the process of an RFID-based ticketing system. It begins with initializing all switches and the display, followed by initializing the RFID reader. The system then reads the RFID tag and checks if the card matches a registered user. If no match is found, it displays "No card found." If the card matches, a buzzer activates once, and passenger details appear on the LCD. The system then checks if the balance is greater than 100. If not, the process stops. If sufficient balance is available, the passenger selects an entry location, scans the RFID tag again for the destination, and selects the destination. The system calculates the total distance travelled and deducts the ticket amount from the balance, completing the process.

6.4 Key Aspects Of The Project

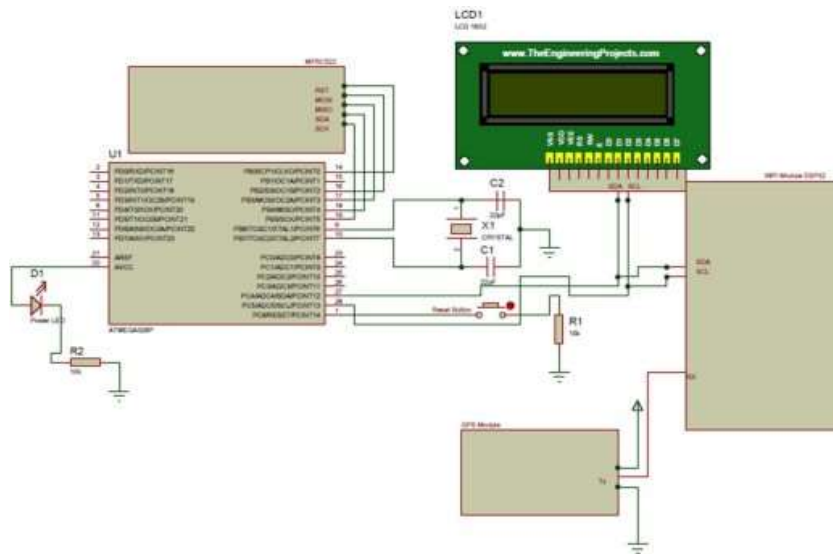
In an IoT-based smart bus ticketing system, passengers can use a card to pay for their bus rides. Before using the card, they need to go through a simple registration and login process. Once registered, passengers can check their card balance, see how many passengers are currently on a bus, view their travel history, and even recharge their card easily through a provided link or platform. This system makes bus travel more convenient and efficient by reducing the need for cash and offering real-time information.

Chapter 7

CIRCUIT DIAGRAM

Chapter 7 – CIRCUIT DIAGRAM

7.1 Circuit Diagram-



7.2 Pin diagram connection-

The ATmega328 acts as the central microcontroller unit (MCU) in this system, coordinating communication and processing data from multiple peripherals. It interfaces with an LCD display via the I2C protocol using the SDA and SCL lines, allowing it to display information such as GPS coordinates or RFID card IDs. An MFRC522 RFID module communicates with the ATmega328 through the SPI interface using pins like MISO, MOSI, SCK, SDA, and RST, enabling the MCU to read unique card identifiers. A GPS module is connected via UART (Rx/Tx), providing continuous location data in the form of NMEA sentences, which the ATmega328 parses to extract latitude, longitude, and time. Additionally, an ESP32 microcontroller, which supports Wi-Fi and Bluetooth, is connected to the ATmega328, typically through GPIO or UART pins, facilitating wireless transmission of data such as RFID events or GPS coordinates to a remote server or other devices. Together, this setup allows the ATmega328 to gather and process inputs from the GPS and RFID modules, display relevant information on the LCD, and communicate wirelessly via the ESP32, creating an integrated system for tracking, identification, and remote data monitoring.

7.3 Pin Assignments-

The ATmega328 interfaces with the MFRC522 RFID module using the

SPI protocol, where its CS/SS pin (Pin 27) connects to the SDA pin of the MFRC522, MOSI (Pin 17) to MOSI, MISO (Pin 18) to MISO, SCK (Pin 19) to SCK, and RST (Pin 16) to RST. For wireless communication, the ATmega328 connects to the ESP32 using multiple GPIO pins, which can be used for data transmission or control signals. The GPS module is connected via UART, with the ATmega328 Tx going to the GPS Rx and the ATmega328 Rx receiving data from the GPS Tx. An I2C-based LCD display is connected to the ATmega328 using the SCL and SDA lines for displaying information. Regarding power supply, the MFRC522 operates at 3.3V, while the other modules, including the ESP32 and LCD, use either 3.3V or 5V depending on their specifications.

Chapter 8

HARDWARE AND COMPONENTS

Chapter 8-Hardware And Components

8.1 Hardware and Components

1.RFID



Figure 4: RFID

Integrated MF RC522

13.56MHz contactless communication card chip. The low-voltage, low-cost, small size of the non-contact card chip to read and write.

Working current : 13 – 26mA / DC 3.3V Standby current : 10 – 13mA / DC 3.3V Sleep current : 80uA

Peak current : 30mA

Working frequency : 13.56MHz

Card reading distance : 0 60mm (Mifare1 card) Here are some details about the MFR522 RFID:

What it is?

The MFR522 is a contactless communication chip that can read and write data. It's designed to be low cost, low power, and compact.

How it works?

The MFR522 controls an antenna that sends and receives data. When an RFID card or tag approaches the antenna, the card's built-in microchip receives energy from the electromagnetic waves emitted by the antenna. The microchip then manipulates the load so that the base station can read the card's data.

Features

The MFR522 supports ISO/IEC 14443 A/MIFARE and NTAG. It also supports rapid CRYPTO1 encryption to validate MIFARE products. The MFR522 can communicate with a host machine via SPI, I2C, or serial UART mode.

Power consumption

The MFR522's continuous current is 13–26 mA, its idle current is 10–13 mA,

and its sleep current is less than 80 A.

Applications

The MFRC522 is a good choice for smart meters and portable handheld devices.

Features of RFID:

1. Integrated MF RC522
2. 13.56MHz contactless communication card chip.
3. The low-voltage, low-cost, small size of the non-contact card chip to read and write.
4. Suitable for Smart meters and portable handheld devices.
5. Advanced modulation and demodulation concept completely integrated into all types of 13.56MHz passive contactless communication methods and protocols.
6. ISO14443A compatible transponder signals.
7. ISO14443A frames and error detection.
8. Supports rapid CRYPTO1 encryption algorithm, terminology validation MIFARE products.
9. MFRC522 support MIFARE series of high-speed non-contact communication, two-way data transmission rates up to 424kbit/s.
10. Low cost, and ideal for user equipment development.
11. The reader and RF card terminal design meet advanced applications development and production needs.
12. Can be directly loaded into the various reader molds, very convenient.

2. Microcontroller-



Figure 5: Microcontroller-ESP32

ESP32 comes with an on-chip 32-bit microcontroller with integrated Wi-Fi + Bluetooth + BLE features that targets a wide range of applications. It is a series of

low-power and low-cost developed by Espresso Systems.

ESP32 is a series of low-cost, low-power system on a chip microcontrollers with integrated Wi-Fi and dual-mode Bluetooth. The ESP32 series employs either a Ten silica Xtensa LX6 microprocessor in both dual-core and single-core variations, an Xtensa LX7 dual-core microprocessor, or a single-core RISC-V microprocessor and includes built-in antenna switches, RF balun, power amplifier, low-noise receive amplifier, filters, and power-management modules. Commonly found either on device specific PCBs or on a range of development boards with GPIO pins and various connectors depending on the model and manufacturer of the board.

ESP32 is created and developed by Espresso Systems, a Chinese company based in Shanghai, and is manufactured by TSMC using their 40 nm process. It is a successor to the ESP8266 microcontroller.

Features of ESP-Wroom-32

- ESP-Wroom-32 contains a low-power Ten silica Xtensa Dual-Core 32-bit LX6 microprocessor at 240 MHz: 994.26 CoreMark; 4.14 CoreMark/MHz
- 448 KB of ROM for booting and core functions.
- 520 KB of on-chip SRAM for data and instructions.
- 4MB of Flash Memory
- 16 KB SRAM in RTC
- Wi-Fi 802.11b/g/n

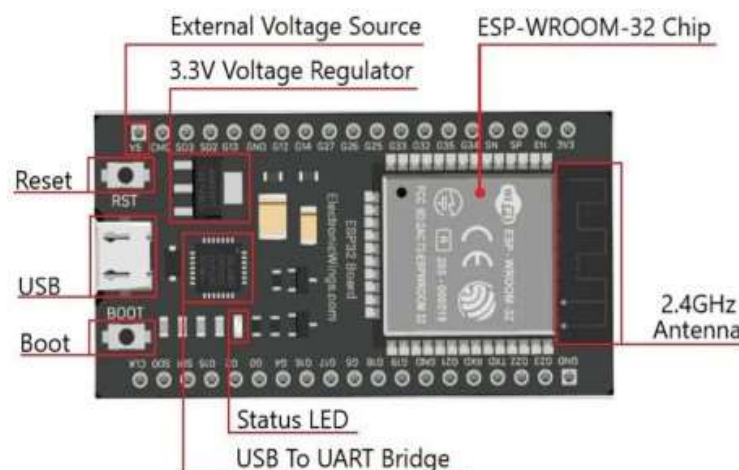


Figure 6: Internal design of ESP-32

ESP-WROOM-32

- ESP-WROOM-32 development board containing Ten silica Xtensa. Dual- Core 32-bit LX6 microprocessor operates at 80 to 240 MHz adjustable clock frequency.
- It comes with 448 KB of ROM, 520 KB of on-chip SRAM, and 4MB of Flash Memory.
- Wireless Connectivity
- On-chip Wi-Fi supports 802.11b/g/n standard
- Operates at 2.4 GHz band, up to 150 Mbps
- It also supports Bluetooth v4.2 BR/EDR and Bluetooth LE specifications this dual mode of Bluetooth makes it even more versatile
- Reset/Boot buttons
- In ESP32 board comes with two main push buttons one is the Reset (RST/EN) button another is the BOOT button.
- The reset button is used to reset the ESP32 Chip.
- The use of the boot button is to enter in boot mode to upload the new sketch or program Power.

ESP32 development board power up from two sources one is from an external 5 Volt source and another from the USB power source.

- The ESP32 Board operates at a 2.3 Volt to 3.6 Volt voltage source and its recommended output current is 600 mA or more.
- Most of the ESP32 development board has an inbuilt 1117 voltage regulator which converts the 5-volt input to 3.3 volts at the output.
- Status Indicator
- In ESP32 development board has one red led status indicator, which shows status while program uploading and serial communication.

Way of Programming for ESP32

The ESP32 supports both C/C++ and Micro Python programming languages that you can use to program your ESP32. Now, let's check the supported ide for both types of languages.

Here's a list of IDEs that supports to program the ESP32 series boards with C/C++.

- ESP-IDF
- Arduino IDE
- VS Code
- Here's a list of IDEs that supports to program the ESP32 series boards with Micro Python.
- Thompy IDE
- PyCharm
- Mu Editor
- up Craft IDE
- VS Code + Piemaker extension

3. 2*16 LCD Display

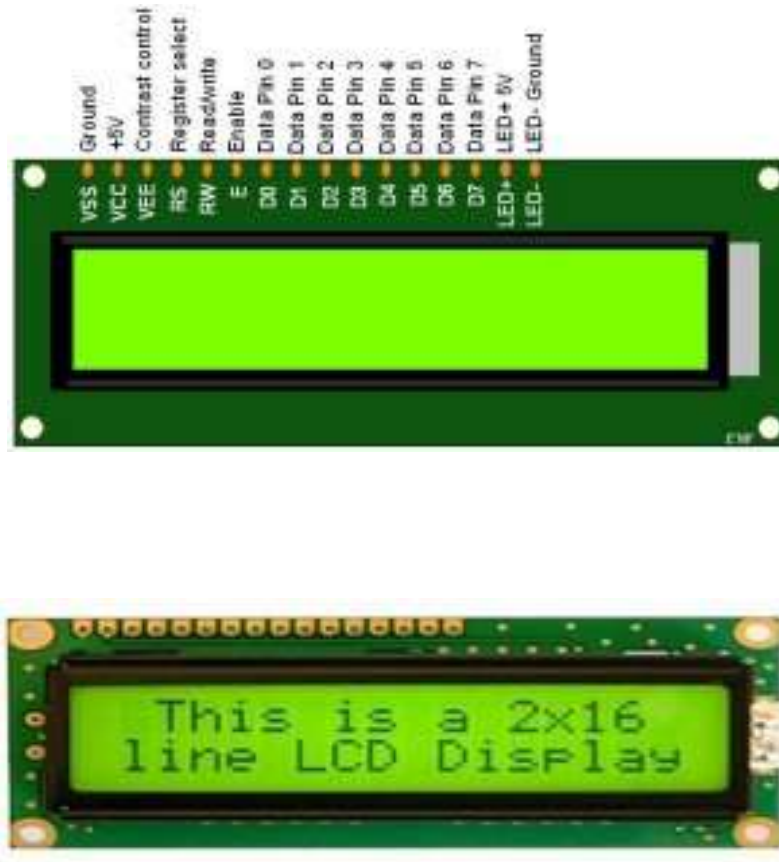


Figure 7: 2*16 LCD display

16 2 LCD is named so because; it has 16 Columns and 2 Rows. There are a lot of combinations available like, 8 1, 8 2, 10 2, 16 1, etc. but the most used one is the 16 2 LCD. So, it will have $(16 \times 2 = 32)$ 32 characters in total and each character will be made of 5×8 Pixel Dots. A Single character with all its Pixels is shown in the below picture.

Now, we know that each character has $(5 \times 8 = 40)$ 40 Pixels and for 32 Characters we will have $(32 \times 40 = 1280)$ 1280 Pixels. Further, the LCD should also be instructed about the Position of the Pixels. Hence it will be a hectic task to handle every- thing with the help of MCU, hence an Interface IC like HD44780 is used, which is mounted on the backside of the LCD Module itself. The function of this IC is to get the Commands and Data from the MCU and process them to display meaningful information onto our LCD Screen. You can learn how to interface an LCD using the above-mentioned links. If you are an advanced programmer and

would like to create your own library for interfacing your Microcontroller with this LCD module then you have to understand the HD44780 IC working and commands which can be found in its datasheet.

Specifications: -

- Operate: 5v DC
- LCD type: characters
- Size: 16x2 (2 rows and 16 characters per row)
- LCD display module with yellow backlight
- Built-in industry standard hd44780 equivalent LCD controller
- Size: 16x2 (2 rows and 16 characters per row)
- LCD display module with yellow backlight
- Built-in industry standard hd44780 equivalent LCD controller

4.GPS



Figure 8: GPS

A GPS module is a compact device used for determining precise location, time, and velocity by connecting to satellites in the Global Positioning System. It provides latitude, longitude, altitude, and time data, typically in NMEA (National Marine Electronics Association) format. Common modules like NEO-6M or Ublox communicate with microcontrollers via UART or I2C interfaces.

They operate on a 3.3V or 5V power supply and include an external antenna for improved signal reception. GPS modules are widely used in navigation systems, vehicle tracking, drones, and IoT applications due to their accuracy and ease of integration.

5.ATMEGA 328 —



Figure 9 : AT MEGA 328

The ATmega328 is a popular microcontroller from the AVR family developed by Microchip Technology. It is widely used in embedded systems, especially in Arduino boards like the Arduino Uno. Below is a detailed overview:

Key Features of ATmega328

1. Architecture: 8-bit AVR RISC-based architecture.
2. Operating Voltage: 1.8V to 5.5V.
3. Clock Speed: Up to 20 MHz
4. Flash Memory: 32 KB (with 0.5 KB used for bootloader).
5. SRAM: 2 KB.
6. EEPROM: 1 KB.
7. GPIO Pins: 23 programmable I/O lines
8. Timers: 2 × 8-bit timers. 1 × 16-bit time

6.Rectifier-



Figure 10 : Rectifier

A rectifier is an electrical device that converts alternating current (AC) to direct current (DC). This process is called rectification, and rectifiers are commonly used in power supply systems and electronic devices.

7. Filter-

A filter is an electrical or electronic circuit designed to allow specific frequency components to pass while attenuating others. Filters are widely used in signal processing, power supplies, and communication systems

Key Parameters

- Cutoff Frequency (f_c): The frequency at which the filter begins to attenuate the signal.



Figure 11: Filter

- Bandwidth (BW): The range of frequencies that the filter passes or rejects.
- Roll-off Rate: The rate at which the filter attenuates frequencies beyond the cutoff.
- Order of the Filter: Higher-order filters provide steeper roll-off rates.
- Q-Factor (Quality Factor): Represents the selectivity or sharpness of a filter's frequency response in band-pass or band-stop filters.

8. Voltage regulator-



Figure 12: Voltage regulator

A voltage regulator is a device that maintains a steady output voltage, even if the input voltage or load changes. It's like a manager for electrical power, ensuring sensitive devices like phones, TVs, and computers receive the right amount of electricity to function properly. Without a voltage regulator, these devices could get damaged or stop working due to power fluctuations.

Chapter 9

RESULT AND DISCUSSION

Chapter 9-RESULT AND DISCUSSION

Result and Analysis

Using the system, we can reduce the conductor effects taken by the conductor during rush. Using these the quality of work increases and improve. When the system is being used the all details of passengers can be recorded easily on MSRTC station. When we use this system then we can analysis easily the passenger frequency to the particular route. As we know our India is on the way of complete digitalization so it's one step from our side improvement in transportation system of India. Using these we can easily avoid frauds and loss.

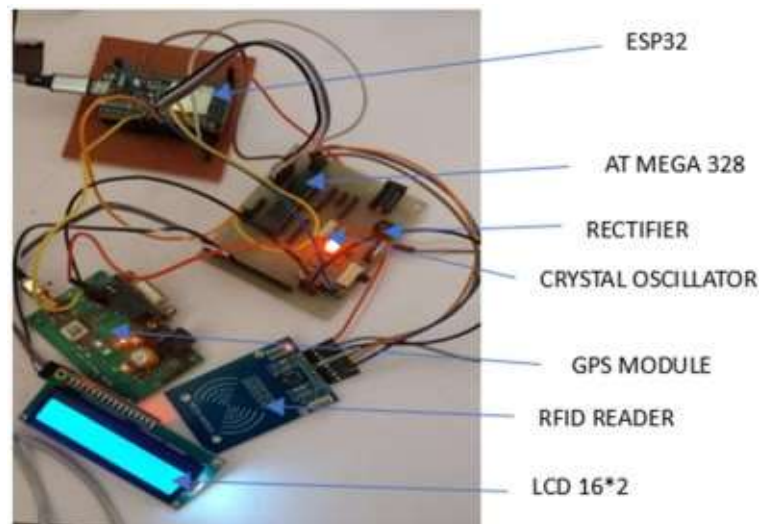


Figure 13: Prototype circuit

An IoT-based smart bus ticketing system modernizes and streamlines public transportation by integrating advanced technology into the ticketing process. Passengers can use a smart card instead of cash or paper tickets for bus rides. To get started, passengers need to register and log in through a dedicated link or online platform. The registration process is simple and allows passengers to create an account to access various features.

One of the key benefits of this system is the ability to check the smart card balance. Passengers can log into their accounts to view their remaining balance and ensure they have enough credit for their trips. If the balance is low, they

can recharge their cards online, making the process quick and hassle-free without requiring physical visits to recharge points.

The system also provides real-time information, allowing passengers to check the number of people currently on a bus. This feature helps passengers make informed decisions, especially during busy hours, by choosing less crowded buses for a more comfortable ride. Additionally, passengers can access their travel history, which includes details such as dates, times, and fares of their past trips. This feature helps them track their expenses and better understand their travel patterns.

When a passenger swipes their smart card to pay for a ride, the fare is automatically deducted from their balance. The system immediately sends a text message to the passenger's mobile phone, confirming the amount deducted. This ensures transparency and keeps passengers informed about their transactions.

Overall, this IoT-based smart bus ticketing system offers a seamless and efficient travel solution. By automating processes such as balance checks, recharges, fare deductions, and real-time updates, it significantly improves the public transportation experience. The system makes bus travel more convenient, transparent, and user-friendly, embracing technology to address traditional challenges in the transportation sector. The performance parameter used for the system evaluation are shown in table.

Comparative analysis

Comparison of Smart Bus Ticketing Systems

System	Year	Performance	Training	Communication	Stability	Ticketing	Efficiency	Multi-Payment Support	Passenger Data Management	Hardware Capability
Mobile Enabled Bus System	2014									
GPS Supported Bus System	2015									
Web-Based Tracking	2016									
Online Bus Ticketing	2018									
IoT-Based System	2020									
RFID-Based Ticketing	2021									
Smart Bus Ticketing	2022									
our IoT-Based System										

Calculating performance rating

For each system, we calculate the performance rating (PR) using a weighted formula:

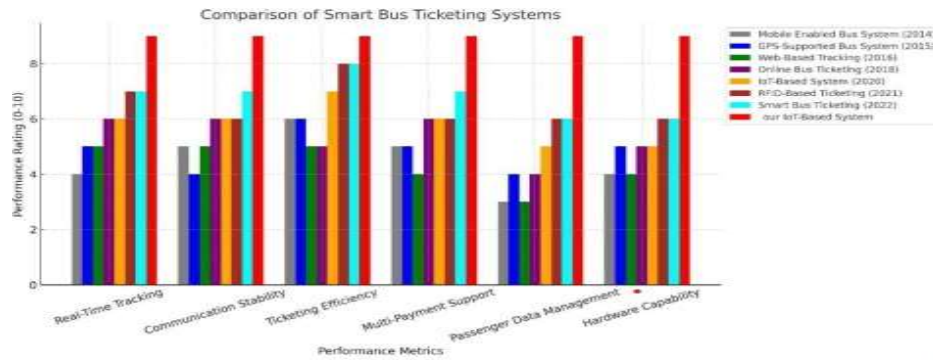


Figure 14: Comparative analysis

$$PR = (W1 * RT) + (W2 * SC) + (W3 * AT) + (W4 * PT) + (W5 * PD) + (W6 * HS)$$

Where:

W1 to W6 are weight factors (important of each metric)

Default weightage:

RT=20 % (W1=0.2)

SC=20 % (W2=0.2)

AT=15 % (W3=0.15)

PT=15 % (W4=0.15)

PD=15 % (W5=0.15)

HS=15 % (W6=0.15)

We assign scores (1-10) to each existing project and our project, then compute their overall performance rating (PR).

Feature	Mobile enables bus tracking (2014)	Our project (IoTbased smart bus ticketing system)
RT	5	10
SC	4	9
AT	3	10
PT	3	9
PD	2	10
HS	4	10

Table 1: Analysis table

Main Dashboard for RFID Based Bus Ticket System

This figure 10.3 shows the admin panel of an RFID-based bus ticketing system, similar to the way modern metro card systems work. When you first land on the screen, there's a login section with the username already filled in as "admin" — probably because this is a demo or default account. The password field is hidden behind asterisks (***) to keep it secure. Once logged in, a friendly "Welcome, admin!" message appears at the top. Below that, a clean, simple menu presents four main options:

Recharge Card – to top up passengers' RFID cards.

Check Live Status – likely showing real-time bus locations or active card usage.

Check History – for reviewing past transactions.

Register Passenger – to add new users to the system.

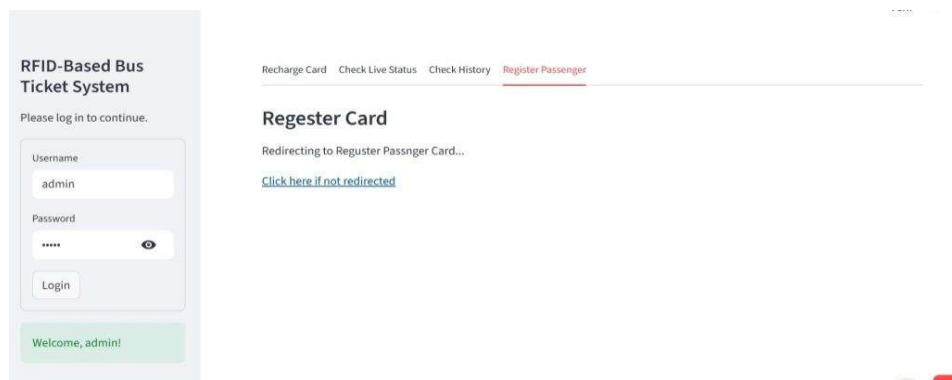


Figure 15: RFID-Based Bus Ticket System: Login and Recharge Portal

Bus Passenger Registration Form

The figure 10.4 shows This is a simple, no-frills digital form for registering new passengers in an RFID-based bus system. Picture a transit employee helping a commuter get their first fare card — this is the form they'd fill out. At the top, there are fields for the passenger's basic info: full name, gender (with "Male" selected by default), age (currently showing "1" as a placeholder), the RFID card number, and the starting balance (set to \$0.00 until funds are added). Below that,

there's a section to upload a photo. Staff can either drag-and-drop or browse for an ID picture. The system supports common formats like JPG and PNG and even allows files up to 200MB — far more than needed for a simple ID. The form finishes with a green “Register” button at the bottom. While it works, there are a few quirks: gender options are limited, the age field shows a placeholder “1,” and there are no fields for email or phone numbers, which could make sending balance alerts or updates tricky. Overall, it gets the job done for quickly issuing RFID cards, but a few tweaks could make the registration process more practical and user-friendly.

BusPassangers Registration Form

Figure 16: Bus Passenger Registration Form

Update Passenger Balance

Figure 10.5 illustrates the interface used by bus operators or ticket agents to top up passengers' RFID fare cards. The design prioritizes transaction speed, a critical requirement during peak hours in crowded transport hubs. The interface is intentionally minimalist: a single prominent “Update Balance” button is positioned for quick access, with no additional input fields, confirmation dialogs, or personal information displays. This design trusts trained staff to process transactions accurately, minimizing cognitive load and interaction steps. By stripping away all nonessential elements, the system ensures that passengers can fund their cards and

continue their journey with minimal delay, thereby improving overall service flow and reducing queue times.

Update Passenger Balance

Enter RFID and additional balance to update the passenger's balance.

Enter RFID:

Enter Additional Balance:

Update Balance

Figure 17: Rapid Top-Up Interface for RFID Fare Cards

Live Bus Passengers

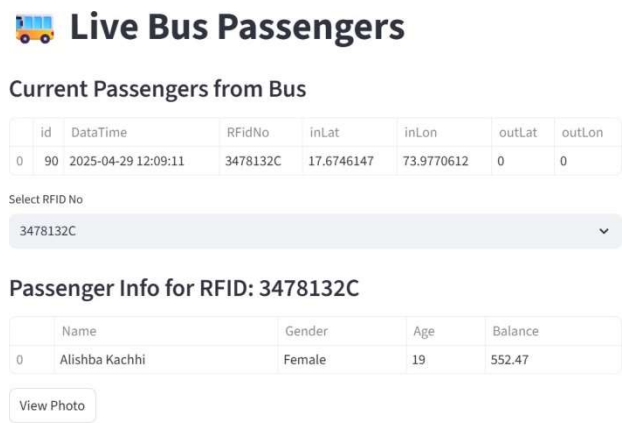


Figure 18: Live Bus Passenger Tracking Dashboard

This figure provides a real-time overview of passengers currently onboard a bus, combining aggregated operational data with **individual** passenger profiles. The upper section displays a table of active riders, including passenger IDs, boarding times, RFID card numbers, and precise GPS coordinates of boarding locations. In the example shown, Passenger ID 0 boarded at 12:09 PM on 29 April 2025 using RFID card 3478132C, with boarding coordinates recorded at latitude 17.6746147 and longitude 73.9770612. The absence of exit coordinates indicates that the passenger remains onboard. The lower section is context-sensitive: selecting a passenger by RFID card number reveals detailed personal information, such as name, age, gender, and current card balance, along with an option to view the registered ID photograph. In this instance, card 3478132C corresponds to Alishba Kachhi, a 19-year-old

female, with a balance of Rs. 552.47. This dual-pane design allows conductors and staff to monitor overall occupancy while also accessing specific passenger details as needed. The interface supports rapid fare verification, rider assistance, and security checks, maintaining operational efficiency even during high passenger volumes along busy routes.

Bus History and Passenger Data Viewer

Bus History and Passengers Data Viewer

Combined Data from BusHistory and BusPassengers

	DateTime	RFidNo	inLocation	outLocation
0	2025-04-29 11:57:07	3478132C	http://maps.google.com/?q=17.67461395,73.97705841	http://maps.goog
1	2025-04-29 11:58:23	D4C9962B	http://maps.google.com/?q=17.67461395,73.97705841	http://maps.goog
2	2025-04-29 12:09:36	D4C9962B	http://maps.google.com/?q=17.67461395,73.97705841	http://maps.goog
3	2025-04-30 09:48:05	D4C9962B	http://maps.google.com/?q=17.67461395,73.97705841	http://maps.goog

Figure 19: Bus History and Passenger Data Viewer

This figure 10.7 shows a detailed log of past bus trips along with passenger information, giving transit operators a clear picture of travel patterns and fares. The table lists four completed journeys, each with the distance traveled and the fare. Interestingly, the first three trips follow a straightforward Rs. 1 per kilometer rate (for example, 47.53 km = Rs. 47.53), while a shorter 6.58 km trip costs Rs. 13.15 — hinting at a minimum fare or a zone-based pricing system. Passenger details highlight Sanskar Herale (Male) as a frequent rider, taking three trips, while Alishba Kachhi (Female) made a single journey. The clean, organized table makes it easy for staff to verify routes, calculate revenue, or quickly address passenger queries. Overall, this historical log effectively tracks trips and fares, showing how transit systems can use simple records to gain meaningful operational insights.

Bus Occupancy Over Time

The Bus Occupancy Over Time line chart in figure 10.8 shows how the number of passengers on a bus changes throughout the day or along its route. The horizontal axis represents time, while the vertical axis indicates the total number of passengers onboard at each moment. As the line rises, it reflects more passengers boarding the bus, and when it falls, it shows passengers alighting. Peaks in the graph indicate periods of maximum occupancy, often corresponding to rush hours or popular stops, while valleys represent quieter times with fewer passengers. This visualization provides valuable insights for bus operators, helping them identify busy periods, optimize scheduling, manage passenger flow, and improve overall efficiency. By leveraging real-time data from RFID fare cards and GPS tracking, the system ensures accurate monitoring of occupancy patterns and supports informed decisions for route planning and fare management.

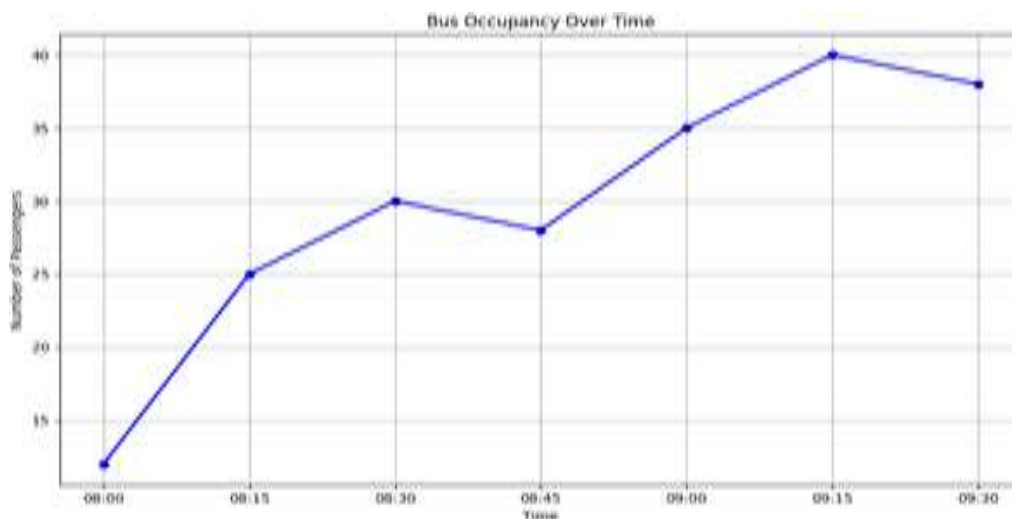


Figure 20: Bus Occupancy Over Time

Number of Passengers per Route Segment

The Number of Passengers per Route Segment bar chart illustrates how passenger traffic varies across different segments of a bus route. Each bar represents a specific segment between two consecutive stops, with the height indicating the number of passengers traveling through that segment. Taller bars show segments with higher passenger loads, highlighting the busiest parts of the route, while shorter bars indicate less crowded segments. This visualization helps bus operators identify which portions of the route experience the most demand, allowing for better resource allocation, such as deploying additional buses during peak segments or adjusting schedules. By analyzing this data from RFID card scans and GPS tracking, the system provides actionable insights to optimize route efficiency, improve passenger comfort, and enhance overall operational planning.

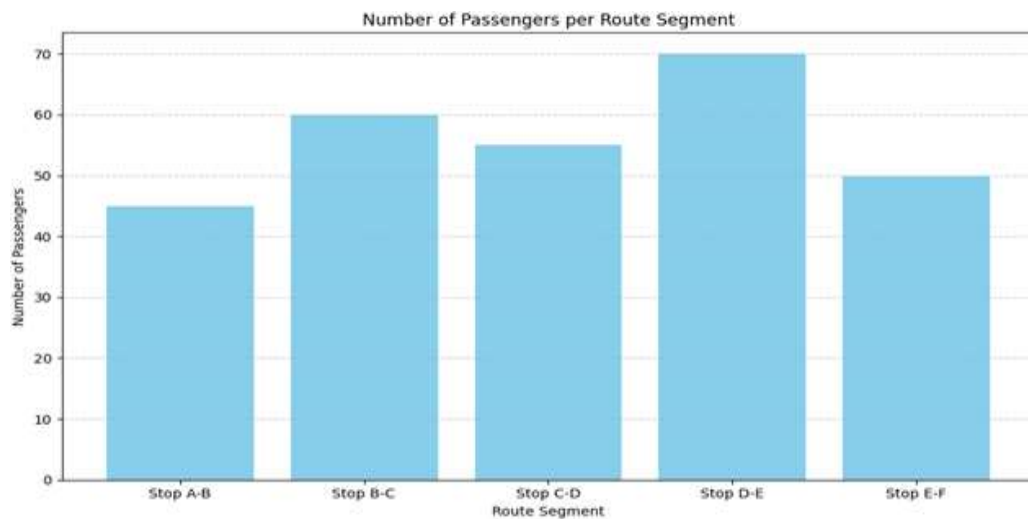


Figure 21: Number of Passengers per Route Segment

Passenger Balance VS Fare Deduction

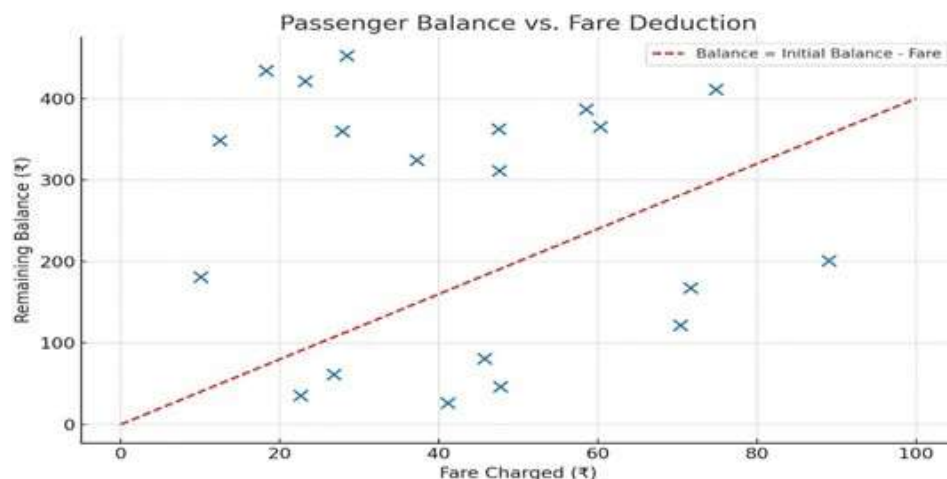


Figure 22: Passenger Balance VS Fare Deduction

The Passenger Balance vs. Fare Deduction graph in figure 10.10 plots each trip's fare against the remaining balance on the passenger's RFID card after payment. Each point shows a passenger's fare on the horizontal axis and their post-deduction balance on the vertical axis, making it easy to see how fares impact available funds. Clusters of points near the lower part of the chart indicate low remaining balances, highlighting potential risk areas for insufficient funds, while higher points represent passengers maintaining healthier balances after travel.

Cumulative Revenue Over Time

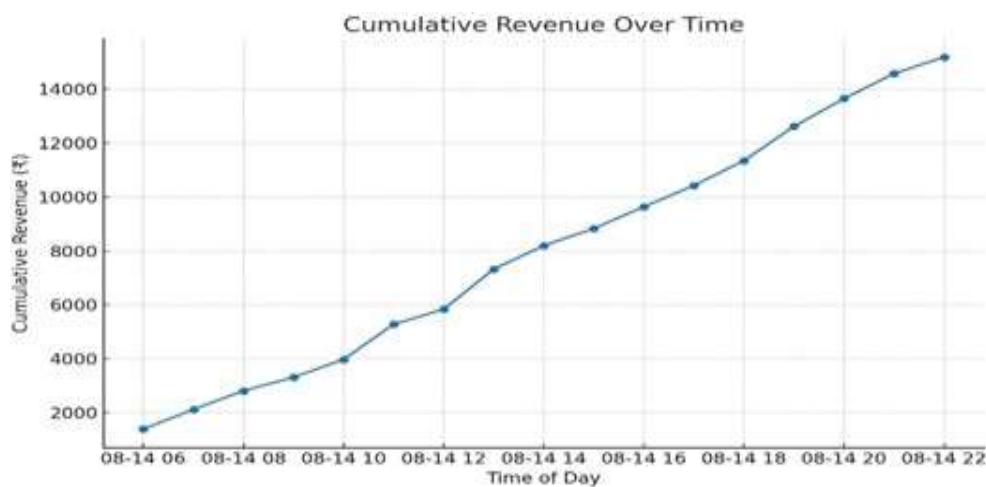


Figure 23: Cumulative Revenue Over Time

The Cumulative Revenue Over Time line chart tracks the total fare revenue collected by the bus service as time progresses. The horizontal axis represents time which could be shown in minutes, hours, or specific bus stops along a route while the vertical axis represents the cumulative sum of fares collected up to that point. The line steadily rises as more passenger's board and pay, with sharper upward slopes during busy periods such as rush hours or high-demand route segments. Flat sections indicate periods with no new fare collection, often when the bus is between stops or in low-demand intervals. This visualization is useful for identifying peak revenue periods, understanding passenger flow patterns, and optimizing scheduling or route planning to maximize efficiency and profitability.

9.1 System Performance Metrics

Metric	Value	Meaning
GPS Distance Accuracy (%)	98.5	How close GPS-calculated distance is to the actual distance traveled.
System Latency (seconds)	1.2	Time between RFID tap and system response.
Transaction Time – Peak Hours (seconds)	3.5	Average time to complete a fare deduction during bus hours.
Transaction Time – Off-Peak Hours (seconds)	2.1	Average fare deduction time during less busy hours.
Cost Savings per Year (Rs.)	Rs. 2,50,000	Annual operational savings (reduced conductors, manual work).
Revenue Increase due to Reduced Fare Evasion (%)	12.5%	Extra revenue from preventing unpaid rides.

Table 2: System Performance and Financial Metrics

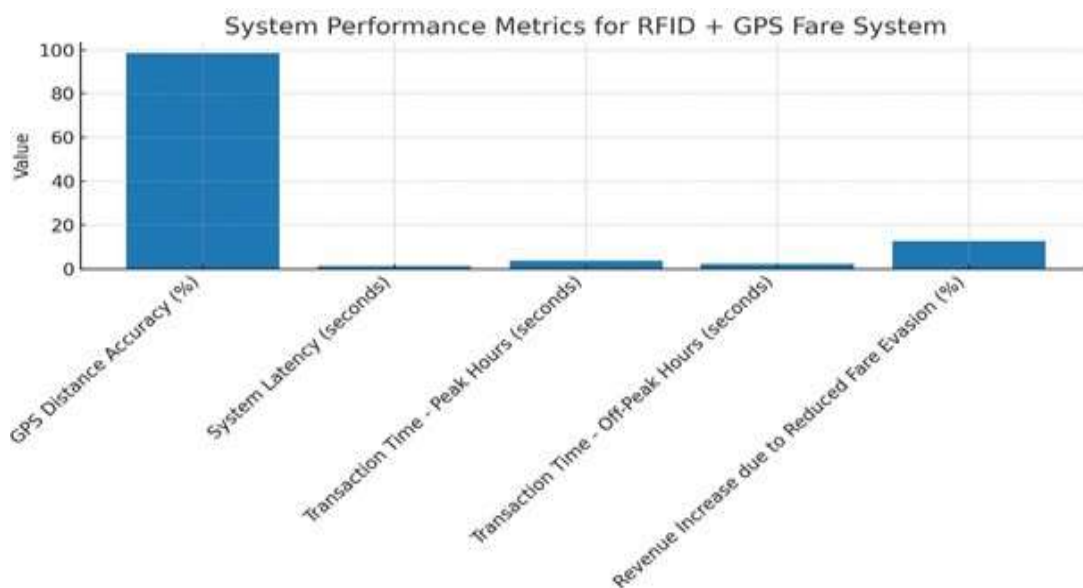


Figure 24: System Performance Metrics

The System Performance Metrics graph in figure 10.12 provides a concise visual summary of how efficiently the RFID and GPS-based fare collection system operates. It shows that the GPS distance accuracy is exceptionally high at around 98.5%, ensuring that fare calculations closely match the actual distance traveled. System latency remains very low at approximately 1.2 seconds, meaning passengers experience minimal delay between tapping their RFID card and receiving a response. Transaction times average 3.5 seconds during peak hours and 2.1 seconds during off-peak periods, reflecting slightly longer processing times when

passenger volume is higher but still maintaining smooth boarding. Additionally, the system delivers a notable 12.5% increase in revenue by reducing fare evasion through accurate passenger tracking and automated fare deduction. Overall, the metrics highlight the system's strengths in accuracy, speed, and financial impact, demonstrating its effectiveness in improving both operational efficiency and revenue assurance for public transportation.

Chapter 10

ADVANTAGES

Chapter 10 – ADVANTAGES

Advantages

Digital ticketing -A digital ticketing system in a smart bus ticketing system allows passengers to book, pay and validate tickets electronically. This system enhances efficiency, reduces frauds, and improve passengers' convenience.

Contact less payment -Contact less payment in smart bus ticketing system enables passengers to pay fares without physical contact using digital methods such as smart cards and mobile wallets.

Fare calculation -A smart bus ticket system calculates fares dynamically based on various factors. The system uses GPS, distance-based pricing, travel zones, and passenger categories to determine the fare amount.

Reusable and rechargeable -A reusable and rechargeable smart ticketing system allows passengers to use a single smart card or digital wallet for multiple trips, reducing waste and enhancing convenience.

Card replacement -When a smart bus card is lost, damaged or expired, a card replacement system ensures passenger can quickly get a new one without losing their balance or passes.

Transaction history -A transaction history feature allows passengers to track their fare payment, travel history and recharges for better financial management and security.

Low-balance alert -A low balance alert notifies passengers when their smart card or mobile wallet balance is running low, ensuring seamless travel without interruption.

Secure encryption -Encryption ensures safe and private transactions in a smart bus ticketing system, protecting passengers' data payment details, and travel history from cyber threats.

Chapter 11

CONCLUSION

Chapter 11 – CONCLUSION

Conclusion

A smart bus ticket is a modern, digital way to pay for bus rides. Instead of using paper tickets or cash, you can use a card, smartphone, or app to pay.

It's convenient because you can top up your balance online, and it makes getting on the bus faster and easier. Plus, it can help reduce paper waste and streamline public transport systems. It is helpful for reduce the trouble of keeping change in pocket.

There is no need of any dependency on the conductor while entering into the bus for collecting the ticket. It also offers a safe way of travelling.

This project proposes a smart digital bus pass and payment system by using android as a platform for development. The digital bus pass system contains features that would help the daily commuters in their day-to-day bus journey not only with ease of use, but also is proved to be a cost-efficient approach in comparison to the traditional system. This work not only contributes to the society but also to the green India campaign eliminating the use of paper. The paper demonstrates the use of android application that eases the hassle of passengers travelling across the city. It also offers a safe way of travelling. This model can be further modified and changed depending on the future needs. Many additional modules and features can be enhanced in the proposed model such as pass category, additional feature of the application, user interface, payment category etc. In this fast-paced growing world, the needs of the people are as ever growing, and it is necessary to keep the technology updated that would match the current requirements. That is why the version of this proposed model has a wide scope of modification and upgradation in the future.

Chapter 12

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Chapter 12 – REFERENCE

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